# Feedback Controls Formula Sheet

Exam 1

## System Modeling and Analysis

### State Space Representation

$\dot{\underline{x}} = [A]\underline{x} + [B]\underline{u}$

$\underline{y} = [C]\underline{x} + [D]\underline{u}$

Mass-Spring-Damper System:

$m\ddot{x} + b\dot{x} + kx = f(t)$

Define the state variables to "kill the dots":

$x = x, \quad v = \dot{x}$

Rewrite the equation of motion as a system of first order differential equations: $m\dot{v} + bv + kx = f(t)$

$$\begin{aligned} \dot{x} &= v \\ \dot{v} &= -\frac{k}{m}x - \frac{b}{m}v + \frac{1}{m}f(t) \end{aligned}$$

In state space form,

$$\underbrace{\begin{bmatrix}\dot{x}\\ \dot{v}\end{bmatrix}}_{\dot{\underline{x}}} = \underbrace{\begin{bmatrix}0 & 1\\ -\frac{k}{m} & -\frac{b}{m}\end{bmatrix}}_{[A]}\underbrace{\begin{bmatrix}x\\ v\end{bmatrix}}_{\underline{x}} + \underbrace{\begin{bmatrix}0\\ \frac{1}{m}\end{bmatrix}}_{[B]}\underbrace{f(t)}_{\underline{u}}$$

$$\underbrace{y}_{\underline{y}} = \underbrace{\begin{bmatrix}1 & 0\end{bmatrix}}_{[C]}\underbrace{\begin{bmatrix}x\\ v\end{bmatrix}}_{\underline{x}} + \underbrace{0}_{[D]}\underbrace{f(t)}_{\underline{u}}$$

### Transfer Function Matrix from State Space

$$\frac{\underline{Y}}{\underline{U}} = [C][sI - A]^{-1}[B] + [D]$$

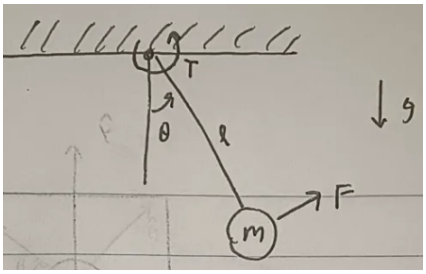
Recall: We can invert a 2x2 matrix using the formula:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad \text{if } ad - bc \neq 0.$$

### Linearization (Jacobian Method)

Nonlinear pendulum: torque input $T = k_\tau \sin(u_1)$ and force input $F = k_F \sin(u_2)$, where $u_1$ and $u_2$ are independent inputs:




The equation of motion is given by:

$$\sum \tau = I\ddot{\theta} \implies m\ell^2\ddot{\theta} = -mg\ell\sin\theta + k_\tau \sin(u_1) + \ell k_F \sin(u_2)$$

Define $\omega = \dot{\theta}$ and the state vector:

$$\underline{x} = \begin{pmatrix}\theta\\ \omega\end{pmatrix}, \quad \underline{u} = \begin{pmatrix}u_1\\ u_2\end{pmatrix}$$

Nonlinear state-space representation:

$$\dot{\underline{x}} = \begin{pmatrix}\omega\\ -\frac{g}{\ell}\sin\theta + \frac{k_\tau}{m\ell^2}\sin(u_1) + \frac{k_F}{m\ell}\sin(u_2)\end{pmatrix} \iff \dot{\underline{x}} = \underline{f}(\underline{x}, \underline{u})$$

$$y = \begin{pmatrix}1 & 0\end{pmatrix}\underline{x} \quad \iff \quad y = \underline{h}(\underline{x}, \underline{u}) \quad \text{(output } y = \theta\text{)}$$

Linearization procedure:

I. Define operating point $(\bar{\underline{x}}, \bar{\underline{u}})$ & incremental variables:

$\underline{x} = \bar{\underline{x}} + \hat{\underline{x}}, \quad \underline{u} = \bar{\underline{u}} + \hat{\underline{u}}, \quad \underline{y} = \bar{\underline{y}} + \hat{\underline{y}}$

Linearize the nonlinear state equation:

$$\underline{f}(\underline{x}, \underline{u}) \approx \underline{f}(\bar{\underline{x}}, \bar{\underline{u}}) + \left.\frac{\partial \underline{f}}{\partial \underline{x}}\right|_{(\bar{\underline{x}}, \bar{\underline{u}})}\hat{\underline{x}} + \left.\frac{\partial \underline{f}}{\partial \underline{u}}\right|_{(\bar{\underline{x}}, \bar{\underline{u}})}\hat{\underline{u}}$$

- State Jacobian $[A]$:

$$[A] = \left.\begin{pmatrix}\frac{\partial f_1}{\partial \theta} & \frac{\partial f_1}{\partial \omega}\\ \frac{\partial f_2}{\partial \theta} & \frac{\partial f_2}{\partial \omega}\end{pmatrix}\right|_{(\bar{\underline{x}}, \bar{\underline{u}})} = \begin{pmatrix}0 & 1\\ -\frac{g}{\ell}\cos\bar{\theta} & 0\end{pmatrix}$$

- Input Jacobian $[B]$:

$$[B] = \left.\begin{pmatrix}\frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2}\\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2}\end{pmatrix}\right|_{(\bar{\underline{x}}, \bar{\underline{u}})} = \begin{pmatrix}0 & 0\\ \frac{k_\tau}{m\ell^2}\cos\bar{u}_1 & \frac{k_F}{m\ell}\cos\bar{u}_2\end{pmatrix}$$

II. Find operating point: $\dot{\bar{\underline{x}}} = \underline{f}(\bar{\underline{x}}, \bar{\underline{u}}) = 0$

$$\dot{\underline{x}}|_{(\bar{\underline{x}}, \bar{\underline{u}})} = \begin{pmatrix}\bar{\omega}\\ -\frac{g}{\ell}\sin\bar{\theta} + \frac{k_\tau}{m\ell^2}\sin(\bar{u_1}) + \frac{k_F}{m\ell}\sin(\bar{u_2})\end{pmatrix} = \begin{pmatrix}0\\ 0\end{pmatrix}$$

Solving these conditions define $(\bar{\underline{x}}, \bar{\underline{u}})$. Leave it symbolic for now.

III. Substitute linearized terms into state equation and apply equilibrium condition:

$(\dot{\bar{\underline{x}}} + \dot{\hat{\underline{x}}}) \approx \underline{f}(\bar{\underline{x}}, \bar{\underline{u}}) + A\hat{\underline{x}} + B\hat{\underline{u}}$

Equilibrium condition $\underline{f}(\bar{\underline{x}}, \bar{\underline{u}}) = 0$ gives:

$\dot{\hat{\underline{x}}} = A\hat{\underline{x}} + B\hat{\underline{u}}$

$$\underbrace{\begin{pmatrix}\dot{\hat{\theta}}\\ \dot{\hat{\omega}}\end{pmatrix}}_{\dot{\hat{x}}} = \underbrace{\begin{pmatrix}0 & 1\\ -\frac{g}{\ell}\cos\bar{\theta} & 0\end{pmatrix}}_{A}\underbrace{\begin{pmatrix}\hat{\theta}\\ \hat{\omega}\end{pmatrix}}_{\hat{x}} + \underbrace{\begin{pmatrix}0 & 0\\ \frac{k_\tau}{m\ell^2}\cos\bar{u}_1 & \frac{k_F}{m\ell}\cos\bar{u}_2\end{pmatrix}}_{B}\underbrace{\begin{pmatrix}\hat{u}_1\\ \hat{u}_2\end{pmatrix}}_{\hat{u}}$$

Output can be linearized in a similar manner:

$$C = \left.\frac{\partial h}{\partial x}\right|_{(\bar{\underline{x}}, \bar{\underline{u}})} = \left.\begin{pmatrix}\frac{\partial h}{\partial \theta} & \frac{\partial h}{\partial \omega}\end{pmatrix}\right|_{(\bar{\underline{x}}, \bar{\underline{u}})} = \begin{pmatrix}1 & 0\end{pmatrix}$$

$$D = \left.\frac{\partial h}{\partial u}\right|_{(\bar{\underline{x}}, \bar{\underline{u}})} = \left.\begin{pmatrix}\frac{\partial h}{\partial u_1} & \frac{\partial h}{\partial u_2}\end{pmatrix}\right|_{(\bar{\underline{x}}, \bar{\underline{u}})} = \begin{pmatrix}0 & 0\end{pmatrix}$$

$$\implies \hat{y} = C\hat{x} + D\hat{u} = \begin{pmatrix}1 & 0\end{pmatrix}\hat{x}$$

Note: For a scalar nonlinear system, linearization is simply a first-order Taylor expansion about the operating point:

$$f(x) \approx f(\bar{x}) + \left.\frac{df}{dx}\right|_{x=\bar{x}}(x - \bar{x})$$

### Laplace Transforms

| | |
|---|---|
| $\delta(t)$ | $1$ |
| $1$ | $1/s$ |
| $t$ | $1/s^2$ |
| $e^{at}$ | $1/(s-a)$ |
| $\dot{f}(t)$ | $sF(s) - f(0^-)$ |
| $\ddot{f}(t)$ | $s^2F(s) - sf(0^-) - \dot{f}(0^-)$ |

### Initial and Final Value Theorems:

$$\boxed{f(0^+) = \lim_{s\to\infty} sF(s)} \qquad \boxed{f(\infty) = \lim_{s\to 0} sF(s)}$$

### Transfer Functions and Block Diagrams

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\text{"Forward Gain"}}{1 + \text{"Loop Gain"}}$$

Series Blocks: $G(s) = G_1(s)G_2(s)$
Parallel Blocks: $G(s) = G_1(s) + G_2(s)$

## System Behavior and Root Locus

### DC Gain and Steady State Error

$$K_{DC} = \lim_{s\to 0} sY(s) = \lim_{s\to 0} G(s) \quad \text{(For unit step input)}$$

$$e_{ss} = \lim_{s\to 0} sE(s), \quad e(t) = r(t) - y(t)$$

### First Order System

$$\tau\dot{y} + y = Ku \leftrightarrow G(s) = \frac{K}{\tau s + 1}$$

$$y(t) = K(1 - e^{-t/\tau}) \quad \text{(For unit step input)}$$

### Second Order System

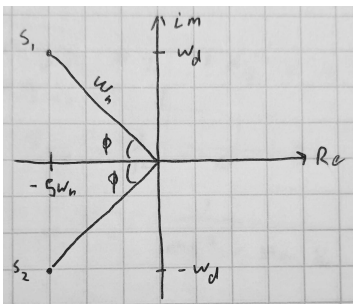
$$\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = \omega_n^2 u \leftrightarrow G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$\sigma = \zeta\omega_n$ (exponential damping rate)

$\omega_d = \omega_n\sqrt{1-\zeta^2}$ (damped frequency)

Poles: $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} = -\sigma \pm \hat{\imath}\omega_d$

$$\phi = \cos^{-1}\left(\frac{\sigma}{|s_{1,2}|}\right) = \cos^{-1}\left(\frac{\zeta\omega_n}{\omega_n}\right) = \cos^{-1}(\zeta)$$



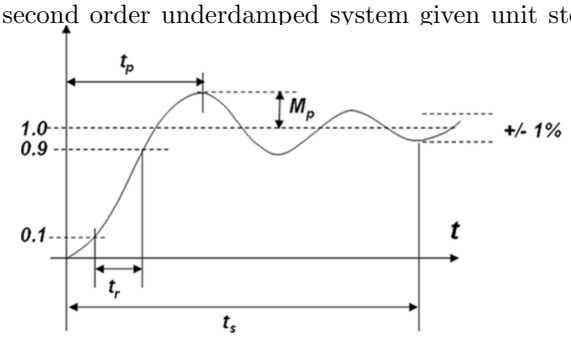

For unit step input, the time response is given by:

$$\implies y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\sin(\omega_d t + \phi)$$

| | |
|---|---|
| Undamped ($\zeta = 0$) | $A\cos(\omega_n t) + B\sin(\omega_n t)$ |
| Underdamped ($0 < \zeta < 1$) | $e^{-\zeta\omega_n t}[A\cos(\omega_d t) + B\sin(\omega_d t)]$ |
| Critically Damped ($\zeta = 1$) | $(C_1 + C_2 t)e^{-\omega_n t}$ |
| Overdamped ($\zeta > 1$) | $C_1 e^{s_1 t} + C_2 e^{s_2 t}$ |

### Time-Domain Specs

For second order underdamped system given unit step input:




| | |
|---|---|
| Rise Time (0–100%) | $t_r = \dfrac{\pi - \phi}{\omega_d} = \dfrac{\pi - \cos^{-1}(\zeta)}{\omega_n\sqrt{1-\zeta^2}}$ |
| Rise Time (10–90%) | $t_r \approx \dfrac{1.8}{\omega_n}$ |
| Peak Time | $t_p = \dfrac{\pi}{\omega_d} = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}$ |
| Max. Overshoot | $M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100\%$ |
| | $\implies \zeta = \dfrac{-\ln(M_p)}{\sqrt{\pi^2 + [\ln(M_p)]^2}}$ |
| Settling Time (2%) | $t_s \approx \dfrac{4}{\zeta\omega_n}$ |
| Settling Time (5%) | $t_s \approx \dfrac{3.6}{\zeta\omega_n}$ |

### Evans Root Locus Rules

$1 + KL_0(s) = 0, \quad K \geq 0$ (Evans Form)

$$\boxed{|KL_0(s)| = 1 \implies K = \frac{1}{|L_0(s)|} \quad \text{(magnitude condition)}}$$

$$\boxed{\angle L_0(s) = (2\ell + 1)\pi, \quad \ell \in \mathbb{Z} \quad \text{(phase condition)}}$$

Rule 0: The root locus is symmetric about the real axis.

Rule 1: Branches of the root locus start at the poles of $L_0(s)$ and end at the zeros of $L_0(s)$ or infinity (Asymptotes).

Rule 2: A point $s$ on the real axis is part of the root locus if $k_0 > 0$ : $N_R$ is odd, $k_0 < 0$ : $N_R$ is even where $N_R$ is the number of poles and zeros to the right of $s$.

Rule 3: All asymptotes intersect at the centroid $\alpha$ with direction $\theta_q$, let relative degree $r$ = number of poles − number of zeros:

$$\boxed{\alpha = \frac{\sum p_i - \sum z_j}{r}} \qquad \boxed{\theta_q = \begin{cases} (2q+1)\pi/r, & k_0 > 0 \\ 2q\pi/r, & k_0 < 0 \end{cases} \quad q = 0, \ldots, r-1}$$